RMNet: Dual-Dimensional Difference Recalibration-Guided CNN-VMamba Synergistic Network for Remote Sensing Image Change Captioning

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***Abstract*—Global surface changes are increasingly monitored using multitemporal remote sensing (RS) technologies. As an emerging technology, change captioning (CC) can organically integrate the location information of changed regions with the semantic analysis of regional attributes to generate natural language descriptions of changes, providing critical support for the intuitive interpretation of RS monitoring results. However, the existing methods have two key limitations: first, single-stream CNNs fail to extract spatiotemporal information sufficiently, which leads to the easy omission of subtle changes, while single-stream transformers suffer from relatively high parameter counts; second, using pixel-level difference information directly causes the model to focus on pseudo-changes induced by illumination or noise, reducing the accuracy of real change characterization and subsequent description. To address these issues, this article proposes recalibration-guided CNN-VMamba synergistic network (RMNet), which is a dual-dimensional difference recalibration (D^3R)-guided CNN-VMamba synergistic network, with two core innovations: 1) a dual-stream architecture adopted that combines the strong local feature extraction capability of CNNs with the powerful global context modeling ability of VMamba, further enhanced by a channelwise spatial window attention mechanism; and 2) a D^3R mechanism that optimizes features by highlighting core changes and suppressing pseudo-changes through dimension-specific enhancement strategies. Extensive experiments on the LEVIR-CC dataset demonstrate significant performance improvements across all evaluation metrics, validating the effectiveness of RMNet in overcoming current limitations in RS CC tasks. The code is available at https://github.com/Jiahuiqu/RMNet**

***Index Terms*—Change captioning (CC), enhanced difference, joint features, remote sensing (RS) images, VMamba.**

I. Introduction

With the continuous development of human society and evolution of natural environments, the Earth's surface is undergoing unprecedented rapid changes. The widespread application of multitemporal remote sensing (RS) imagery has provided crucial data support to study these surface transformations. In contrast to conventional change detection [1], [2], [3], [4], [5], change captioning (CC) [6], [7] generates natural language descriptions of altered areas, not only localizing changed regions but also interpreting their attribute characteristics and mutual relationships, as illustrated in Fig. 1 This technology overcomes the inherent constraints of traditional change detection methods that produce binary outputs lacking semantic interpretability and produces more cognitively compatible analysis results that better align with human interpretation patterns. Therefore, once proposed, CC technology has been widely applied in critical applications, including disaster monitoring, land surveys, and environmental assessments.

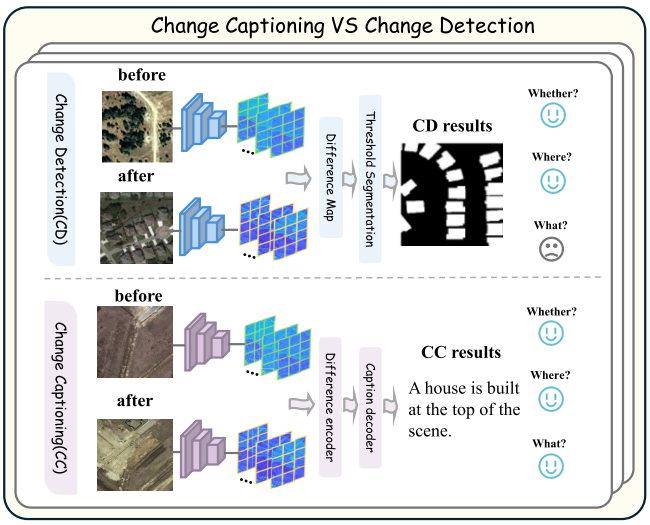


Fig. 1. Comparison of change detection and CC.

Unlike conventional vision tasks, such as scene classification [8], [9], [10], semantic segmentation [11], [12], and image fusion [13], [14], [15], image CC represents a cross-modal task that integrates both visual and language. The processing line consists of two key stages: visual analysis and language generation. In the visual analysis stage, the model must

Received 30 December 2025; revised 20 January 2026; accepted 22 January 2026. Date of publication 26 January 2026; date of current version 13 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62471359, in part by the Key Research and Development Program of Shaanxi under Grant 2025SF-YBXM-513, in part by the Young Talent Fund of Association for Science and Technology in Shaanxi under Grant 20230117 and Grant 20250133, and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX25084. *(Corresponding author: Wenqian Dong.)*

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Digital Object Identifier 10.1109/TGRS.2026.3658213

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precisely localize the changes in the target regions and subsequently identify the change types, including typical change patterns, such as the appearance, disappearance, or attribute modification of objects; in the language generation stage, the model transforms the extracted visual features into natural language that accurately convey semantic change information and comply with grammatical rules and linguistic conventions. Jhamtani and Berg-Kirkpatrick [6] first introduced the CC task and proposed a Siamese CNN-RNN architecture to describe difference features between similar images. To effectively distinguish distractors from actual changes, Park et al. [16] developed a dual dynamic attention (DUDA) model, which localized changes and described them via natural language, significantly improving captioning accuracy and robustness. Chouaf et al. [17] and Hoxha et al. [18] were among the first to explore CC on small-scale RS datasets. Chang and Ghamisi [19] proposed an attentive changes-to-captions (Chg2Caps) network for accurate bi-temporal RS image change descriptions, which sequentially extracted features via a Siamese CNN, located change-related key features and generated image embeddings using an attentive encoder with a hierarchical self-attention block. To tackle insufficient joint spatial–temporal modeling in RSICC, Liu et al. [20] proposed the RSCaMa model. It used ViT as the feature extractor, achieving efficient joint spatial–temporal modeling via multiple CaMa layers. However, the current mainstream methods for RS image CC (RSICC) still have some limitations.

To generate accurate descriptions, CC must encompass both the attributes of changed objects and their positional information within the image's global context, necessitating feature extraction that simultaneously captures local details and global semantics. However, existing methods predominantly rely on single-stream CNN or transformer architectures: CNNs are constrained by local receptive fields and struggle to capture long-range dependencies; while transformers are effective in modeling global context, they exhibit limited sensitivity to fine-grained local variations and incur high computational complexity. Therefore, there is an urgent need to develop a dual-stream architecture capable of extracting comprehensive features while maintaining parameter efficiency. Furthermore, in the difference encoding stage, the existing approaches typically rely on unprocessed pixel-level difference features, which fail to adequately distinguish pseudo-changes caused by factors, such as illumination, noise, and seasonal variations from genuine land-surface changes, thereby further compromising the reliability of generated descriptions and increasing the risk of erroneous judgments.

Mamba [21] based on state-space models (SSMs) has provided a promising direction, offering linear computational complexity for efficient global dependency modeling and laying a technical foundation for constructing an efficient dual-stream collaborative feature extractor. To this end, we propose a dual-dimensional difference recalibration-guided CNN-VMamba synergistic network (RMNet), in which a dual-stream joint (DSJ) feature extraction architecture cooperates with a difference-aware refinement mechanism, enabling both local–global semantic fusion and robust pseudo-change suppression for accurate CC. Specifically, we design a DSJ feature extraction and fusion module. This module adopts a dual-stream parallel architecture in the feature extraction stage, with ResNet extracting local fine-grained features and VMamba modeling global semantics, and combines local–global feature interaction with channelwise spatial window attention to fuse and screen features for comprehensive temporal-phase information. Dual-dimensional difference recalibration (D^3R) module is developed to optimize differential features of bi-temporal images, strengthening real change regions (spatial dimension) and activating change-related channels (channel dimension) to suppress pseudo-changes. Finally, progressive visual-linguistic alignment is executed in the transformer decoder, and cross-modal attention enables accurate visual-language mapping to generate descriptions with spatial interpretability and semantic accuracy. The main contributions of our work are summarized as follows.

1) We propose RMNet, which builds a novel technical paradigm for RSICC via collaborative DSJ feature extraction and fusion and dual-dimensional differential feature enhancement, enabling in-depth mining and accurate expression of change information.
2) The DSJ module synergistically integrates the advantages of two types of networks, fully covering feature information from local to global scales. Additionally, it embeds a learnable channelwise spatial window attention submodule to perform collaborative optimization, enhancing the effectiveness of feature representation.
3) We design the D^3R module, which relies on the synergistic effect of D^3R to eliminate pseudo-changes and focus on regions with real changes, achieving high-quality differential feature optimization.

II. Related Work

A. State-Space Model

SSM [22] is a powerful tool for modeling dynamic systems. By introducing intermediate state variables, they can effectively capture the temporal evolution of systems and have been widely applied in fields, such as signal processing and time-series analysis. Inspired by SSM, the structured state-space sequence (S4) model further improves the traditional SSM. This model maps a 1-D function or sequence $x(t)$ to a response $y(t)$ through a hidden state $h(t)$, which can be mathematically expressed as follows:

$$h(t)' = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) \tag{2}$$

where $x(t)$ denotes the input sequence, $h(t)$ represents the hidden state that facilitates mapping $x(t)$ to $y(t)$, $\mathbf{A}$ denotes the state transition matrix, and $\mathbf{B}$ and $\mathbf{C}$ represent the matrices that map inputs to states and states to outputs, respectively.

In deep learning, we typically deal with discrete sequential data (such as image pixels or time series), necessitating the discretization of continuous signals. Specifically, this involves transforming continuous-time parameters $\mathbf{A}$ and $\mathbf{B}$ into discrete-time parameters $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ using methods like zero-order hold (ZOH)

$$\overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \tag{3}$$

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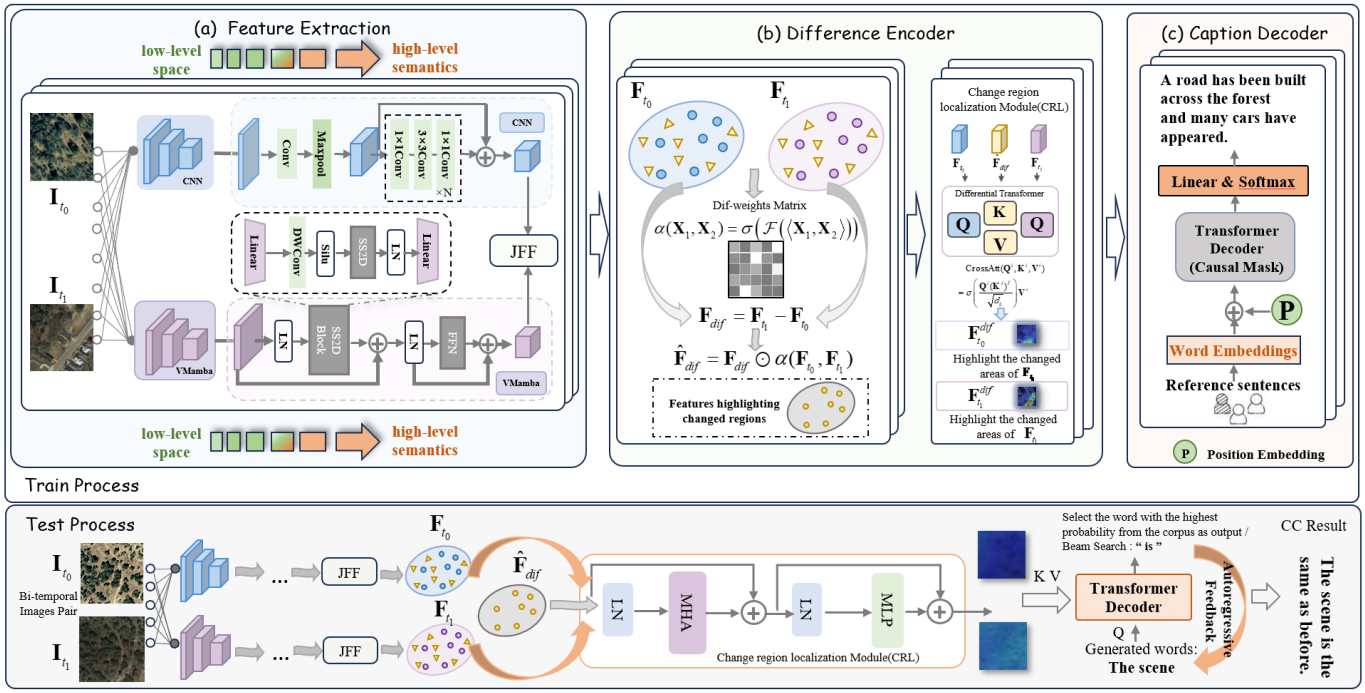


Fig. 2. Overall architecture of RMNet. The feature extraction stage extracts and fuses local and global features using ResNet and VMamba; the difference encoding stage performs dual-dimensional enhancement of pixel-level differences and localizes change regions via cross-attention; finally, the caption decoding stage employs a transformer-based decoder to map the refined features into sentences. (a) Feature extraction. (b) Difference encoder. (c) Caption decoder.

$$\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - I\right)\cdot\Delta\mathbf{B}. \tag{4}$$

After discretization, formulas (1) and (2) can be expressed as follows:

$$h(t) = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \tag{5}$$

$$y(t) = \mathbf{C}h(t). \tag{6}$$

SSM is incapable of performing targeted reasoning for different inputs. In response to this limitation, Mamba [21] has devised a straightforward selection mechanism. By parameterizing the inputs to SSM, the model can selectively process information, thereby focusing on or disregarding specific inputs and enhancing the dynamic interactions between sequences. Moreover, Mamba has proposed a hardware-aware algorithm. Instead of using convolution, this algorithm uses a parallel scanning algorithm for the recurrent computations of the model, ensuring that Mamba maintains linear growth in both memory and computational costs.

B. VMamba

In current visual tasks, traditional architectures face the core contradiction between global information capture efficiency and the adaptation to the nonsequential nature of visual data. By integrating the efficiency of SSMs with the characteristics of visual tasks, VMamba [23] significantly reduces computational costs while ensuring global information capture capability, making it a preferred architecture that balances performance and efficiency.

The core functional module of VMamba is the 2-D selective scanning (SS2D) module, which directly determines the architecture's adaptability to visual data and the efficiency of global information interaction. The scanning operation of traditional SSMs (e.g., S6) is temporally sequential, making it suitable for sequential data in the NLP field. However, visual data are inherently nonsequential and contain multidimensional spatial information, such as local textures and global structures. Direct application of traditional SSMs easily leads to the loss of spatial information. To address this issue, the SS2D module is designed with the following process: first, it flattens the input feature map into 1-D sequences along four complementary 2-D traversal paths, ensuring coverage of all spatial directions; then, it performs selective scanning on the sequences through four parallel S6 modules to dynamically filter key information and suppress redundancy; finally, it reshapes the sequences back into feature maps in the reverse order of the original traversal paths and fuses them. This enables each pixel to integrate global pixel information from multiple directions, efficiently constructing a global receptive field in 2-D space.

Relying on the global receptive field constructed by the SS2D module, VMamba can fully integrate visual information from local to global scales. In tasks, such as image classification and object detection, it can not only capture detailed texture features but also grasp the correlation of overall structures, effectively improving the accuracy of task performance.

C. Change Captioning

Early image captioning techniques primarily included template-based methods [24] and retrieval-based methods [25], [26]. Template-based methods identify ground objects and automatically populate predefined linguistic templates with

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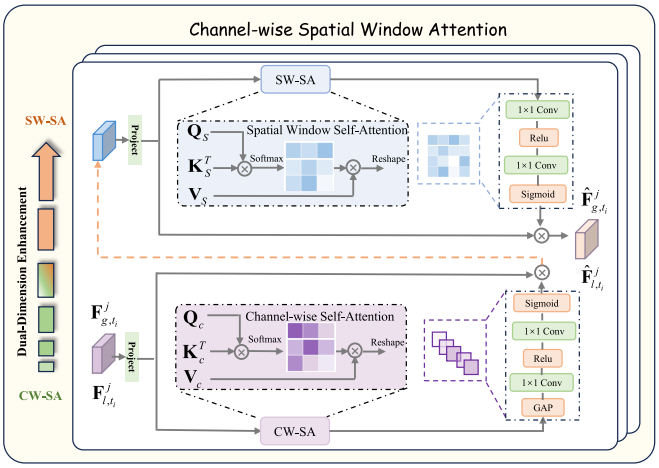


Fig. 3. Channelwise spatial window attention submodule.

detected semantic labels to generate structured descriptive text. Retrieval-based RS image captioning relies on large-scale annotated databases, where the input image is matched with the most similar sample in the library, and its associated prestored caption is directly output.

Currently, the encoder–decoder architecture [27] has emerged as the predominant paradigm for image CC tasks. This framework employs a Siamese network to extract deep features from bi-temporal image pairs separately, followed by a decoder that maps the fused temporal features into natural language descriptions. To enhance model performance, researchers have pursued methodological innovations across multiple dimensions, including feature fusion mechanisms, attention optimization, and language modeling [20], [28]. Addressing the challenge where viewpoint variations obscure meaningful semantic differences, Shi et al. [29] designed a novel visual encoder to disentangle viewpoint changes from semantic changes. They further introduced a reinforcement learning-based attention refinement method to mimic human visual attention and enhance description accuracy. He et al. [30] proposed a gradient-consistent multiscale feature extraction method for cross-modal weak alignment, which provides insights into alleviating pseudo-changes in CC tasks. Additionally, Tu et al. [31] proposed a viewpoint-adaptive representation disentanglement network. This approach leveraged position-embedded representation learning and invariant representation disentanglement to adapt to viewpoint variations.

While the aforementioned methods are implemented on synthetic scenes, real-world RS images present significantly greater challenges for bi-temporal CC. Compared to synthetic data, real RS imagery covers broader spatial scales, encompasses more diverse land-cover types, and exhibits finer-grained object characteristics—all of which substantially increase the complexity of generating accurate change descriptions. Liu et al. [32] constructed LEVIR-CC, a large-scale RS image dataset for CC, and proposed a dual-branch transformer encoder that leverages bi-temporal feature differences as keys to enhance the image features of each temporal instance. To address the potential ambiguity in change descriptions caused by coupled processing of image pairs, Liu et al. [33] proposed a task decoupling framework, which used an image-level classifier for binary classification and a feature-level encoder for feature extraction, then feeding prompts and visual features into a frozen LLM for language generation. Shi et al. [34] argued that the complementary nature of change regions and deep semantic information could further enhance model performance and robustness. Therefore, they proposed a multitask framework to simultaneously address both change detection and CC in RS images. Liu et al. [35] proposed an interactive change agent, which can follow user instructions to achieve comprehensive change interpretation and in-depth analysis. Zhu et al. [36] introduced a knowledge-semantics-guided method, which leveraged latent knowledge from foundation models to reduce annotation reliance and used change detection to provide pixel-level semantic guidance.

III. Methodology

A. Architecture Overview

Given a bi-temporal RS image pair $(\mathbf{I}_{t_0}, \mathbf{I}_{t_1})$, both images are encoded to extract the corresponding features $\mathbf{F}_{t_0}, \mathbf{F}_{t_1}$. A pixel-level difference operation is then applied to obtain the differential feature ($\Delta\mathbf{F} = |\mathbf{F}_{t_1} - \mathbf{F}_{t_0}|$), which is further fused with natural language semantic information. Through this process, dynamic change information $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ (where c_i denotes the ith category of change features) is transformed into natural language descriptions $\mathcal{S} = \{s_1, s_2, \ldots, s_m\}$, achieving an advancement from change detection to CC. Traditional CC methods typically employ pretrained CNN backbone networks or ViT models to extract features $\mathbf{F}_{t_0}$ and $\mathbf{F}_{t_1}$ from bi-temporal RS images $\mathbf{I}_{t_0}$ and $\mathbf{I}_{t_1}$, respectively. Subsequently, without additional optimization, pixel-level numerical differences are directly calculated for this batch of initially extracted features to construct change representations. Finally, the image features enhanced by difference information are fed into LSTM/GRU-based RNNs or transformer decoders to generate natural language descriptions.

However, previous methods usually adopt single-stream CNN or transformer architectures for feature extraction, which may lead to insufficient feature representation and high computational complexity, thus affect the accuracy and efficiency of CC. Additionally, they directly model the difference information obtained through simple subtraction operations, which makes the generated difference features highly sensitive to noise and unable to fully capture real change information. Therefore, this article proposes RMNet integrating the DSJ module and the D^{3}R module, which are used to address the issues of insufficient temporal-phase feature extraction and inaccurate change information representation, respectively. As illustrated in Fig. 2, during the joint feature extraction stage, the DSJ module constructs a dual-stream architecture with two independent ResNet–VMamba pathways, enabling the simultaneous capture of fine-grained spatial features and the exploration of global semantic correlations in images. Meanwhile, a channelwise spatial window attention mechanism is designed to perform precise screening and optimization of

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the initially extracted image features. In the difference information refinement (DIR) stage, aiming at the pseudo-change interference introduced by pixel-level differences, D^{3}R module achieves feature optimization by highlighting core change regions and suppressing pseudo-changes across both spatial and channel dimensions through local feature enhancement and feature channel activation strategies. This provides high-quality feature support for the subsequent CC task. Finally, a transformer-based decoder is employed to generate natural language descriptions that accurately convey both the type and location of changes in a manner consistent with human cognitive habits. This process of the proposed method can be formulated as follows:

$$\begin{aligned}\mathbf{S}_t &= \mathcal{F}_{\text{RMNet}}\left(\mathbf{I}_{t_0}, \mathbf{I}_{t_1}\right) \\ &= \mathcal{F}_{\text{TD}}\left(\mathcal{F}_{\text{D}^3\text{R}}\left(\mathcal{F}_{\text{DSJ}}\left(\mathbf{I}_{t_0}, \mathbf{I}_{t_1}\right)\right)\right)\end{aligned} \tag{7}$$

where $\mathbf{S}_t$ represents the generation result of the change caption, $\mathbf{I}_{t_0}$ and $\mathbf{I}_{t_1}$ denote the two original images, respectively, $\mathcal{F}_{\text{RMNet}}(\cdot)$ represents the proposed CC method, $\mathcal{F}_{\text{DSJ}}(\cdot)$ denotes the dual-stream joint feature extraction and fusion module, $\mathcal{F}_{\text{D}^3\text{R}}(\cdot)$ refers to the dual-dimensional difference recalibration module, and $\mathcal{F}_{\text{TD}}(\cdot)$ represents a standard transformer decoder used to generate the change caption.

B. DSJ Feature Extraction and Fusion Module

To address the issue of insufficient feature extraction in previous change description methods, we design a DSJ feature extraction and fusion module for RS images. This module leverages the local receptive fields of CNN to capture local texture and fine-grained details of images and utilizes the selective state space mechanism of VMamba to model global long-range dependencies and explore global semantics. On this basis, we also design a channelwise spatial window attention submodule to optimize the initially extracted features. This mechanism can fuse local and global features, screen key information, and ultimately achieve comprehensive and high-quality feature characterization of bi-temporal RS images. The process can be formulated as follows:

$$\begin{aligned}\left(\mathbf{F}_{t_0}^{j}, \mathbf{F}_{t_1}^{j}\right) &= \mathcal{F}_{\text{DSJ}}\left(\mathbf{I}_{t_0}, \mathbf{I}_{t_1}\right) \\ &= \mathcal{F}_{\text{JFF}}\left(\mathcal{F}_{\text{JFE}}\left(\mathbf{I}_{t_0}, \mathbf{I}_{t_1}\right)\right)\end{aligned} \tag{8}$$

where $\mathbf{F}_{t_0}$ and $\mathbf{F}_{t_1}$ represent the bi-temporal image features that have undergone complete processing via the DSJ module, $\mathcal{F}_{\text{JFE}}(\cdot)$ denotes the joint feature extraction, and $\mathcal{F}_{\text{JFF}}(\cdot)$ denotes the joint feature fusion.

1) Joint Feature Extraction: For each temporal image $\mathbf{I}_{t_i} \in \mathbb{R}^{H\times W\times 3}[i \in (0,1)]$, two complementary branches are constructed. To efficiently capture the spatial detail features of images while preserving the spatial correlation of the original images, we adopt the ResNet-101 model pretrained on the SSL4EO-S12 [37] dataset and select its first j stages as the local detail extractor. The core operation can be expressed as follows:

$$\mathbf{F}_{l,t_i}^{0} = \text{Maxpool}_{3\times 3}\left(\text{RELU}\left(\text{BN}\left(\text{Conv}_{7\times 7}\left(\mathbf{I}_{t_i}\right)\right)\right)\right) \tag{9}$$

$$\begin{aligned}\mathbf{F}_{l,t_i}^{j} = \text{Relu}\Big(\text{BN}\Big(&\text{Conv}_{1\times 1}\Big(\text{Conv}_{3\times 3}\Big(\text{Conv}_{1\times 1}\Big(\mathbf{F}_{l,t_i}^{j-1}\Big)\Big)\Big)\Big) \\ &+ \text{Conv}_{1\times 1}\Big(\mathbf{F}_{l,t_i}^{j-1}\Big)\Big)\end{aligned} \tag{10}$$

where $\mathbf{F}_{l,t_i}^{j} \in \mathbb{R}^{H/(2^{j+1})\times W/(2^{j+1})\times C'}$ denotes the local feature map generated by the jth stage $[j \in (1,2,3,4)]$ of the CNN branch for the t_ith $[i \in (0,1)]$ temporal image. Stages 1–4 output feature maps with downsampling rates of 4×, 8×, 16×, and 32×, respectively. These feature maps preserve detailed spatial information while progressively abstracting higher level semantic features.

The global feature extraction branch is built upon the VMamba architecture, which achieves efficient long-range dependency modeling through an innovative SSM. The architecture of VMamba shares certain similarities with that of ResNet. Its operations can be expressed by formulas as follows:

$$\mathbf{F}_{g,t_i}^{0} = \text{Conv}_{4\times 4}\left(\mathbf{I}_{t_i}\right) \tag{11}$$

$$\mathbf{F}_{g,t_i}^{1} = \text{VSSBlock}\left(\mathbf{F}_{g,t_i}^{0}\right) \tag{12}$$

$$\mathbf{F}_{g,t_i}^{j} = \text{VSSBlock}\left(\text{downsampling}\left(\mathbf{F}_{g,t_i}^{j-1}\right)\right) \tag{13}$$

where $\mathbf{F}_{g,t_i}^{j} \in \mathbb{R}^{H/(2^{j+1})\times W/(2^{j+1})\times C''}$ denotes the global feature map generated by the jth stage $[j \in (2,3,4)]$ of the Mamba branch for the ith temporal image $[i \in (0,1)]$. These feature maps cover the global information of the image and can effectively capture the key features of the image at the overall level.

2) Joint Feature Fusion: To accurately screen the image features initially extracted by ResNet and VMamba, we design a channelwise spatial window attention submodule. This submodule constructs a dual-dimensional optimization mechanism: in the channel dimension, it selects feature channels that are highly correlated with change semantics via a dynamic gating strategy and eliminates irrelevant redundant channels; in the spatial dimension, it enhances the feature response intensity of core change regions based on window-based local computation, thereby achieving precise purification of the original features and suppression of redundant information. As illustrated in Fig. 3, the core inputs of this submodule are two types of features: one is the global feature $\mathbf{F}_{g,t_i}^{j}$ from the j stages of VMamba and the other is the local feature $\mathbf{F}_{l,t_i}^{j}$ from the j stages of ResNet. For these two types of input features, the submodule adopts a unified processing flow of "channel-dimensional optimization first, followed by spatial window enhancement." After the two types of features are filtered and enhanced, feature concatenation is performed in the channel dimension, ultimately generating fused features that integrate global semantic information, local detailed features, and strong discriminative capability.

Specifically, for the channelwise self-attention (CW-SA) mechanism, the query matrix ($\mathbf{Q}_{g_c}$, $\mathbf{Q}_{l_c}$), key matrix ($\mathbf{K}_{g_c}$, $\mathbf{K}_{l_c}$), and value matrix ($\mathbf{V}_{g_c}$, $\mathbf{V}_{l_c}$) in the channel dimension are constructed from the global feature and the local feature, respectively. The attention weights between channels are obtained by first performing a dot product operation between the transposed key matrix and the query matrix, then activating the result through the Softmax function, and finally

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multiplying it with the value matrix

$$\mathbf{A}_{g_c} = \text{Softmax}\left(\frac{\mathbf{Q}_{g_c}\mathbf{K}_{g_c}^T}{\sqrt{d_{g_c}}}\right)\mathbf{V}_{g_c} \tag{14}$$

$$\mathbf{A}_{l_c} = \text{Softmax}\left(\frac{\mathbf{Q}_{l_c}\mathbf{K}_{l_c}^T}{\sqrt{d_{l_c}}}\right)\mathbf{V}_{l_c} \tag{15}$$

where d_{g_c} and d_{l_c} represent the channel feature dimensions, and this weight matrix quantitatively characterizes the association importance between the features of different channels.

After undergoing a reshape operation, the feature representations processed by channel attention are obtained. To generate the gating coefficients $\mathbf{G}_{g_c}$ and $\mathbf{G}_{l_c}$ in the channel dimension, the aforementioned features need to go through a global average pooling (GAP) operation in sequence to aggregate global channel information, followed by a 1×1 convolution for dimension transformation, a ReLU activation function to introduce nonlinear transformation, another 1×1 convolution to restore the feature dimension, and finally a mapping through the sigmoid function to obtain the channel gating coefficients $\mathbf{G}_{g_c}$ and $\mathbf{G}_{l_c}$

$$\langle \mathbf{G}_{g_c}; \mathbf{G}_{l_c} \rangle = \sigma\left(\text{Conv}_{1\times1}\left(\text{Relu}\left(\text{Conv}_{1\times1}\left(\text{GAP}\left(\langle \mathbf{A}_{g_c}; \mathbf{A}_{l_c} \rangle\right)\right)\right)\right)\right) \tag{16}$$

where σ denotes the sigmoid function. We then perform elementwise multiplication between the gating coefficient $\mathbf{G}_{g_c}$ and the original global features $\mathbf{F}_{g,t_i}^j$ and between the gating coefficient $\mathbf{G}_{l_c}$ and the original local features $\mathbf{F}_{l,t_i}^j$, respectively, yielding local features and global features enhanced by the CW-SA mechanism.

For the spatial window self-attention (SW-SA) mechanism, to accurately capture the correlation information in the spatial dimension, query($\mathbf{Q}_{g_s}$, $\mathbf{Q}_{l_s}$), key($\mathbf{K}_{g_s}$, $\mathbf{K}_{l_s}$), and value($\mathbf{V}_{g_s}$, $\mathbf{V}_{l_s}$) in the spatial dimension are constructed specifically from the local features and global features enhanced by the CW-SA mechanism. Since directly calculating global spatial attention will lead to high complexity, we design a spatial window partitioning strategy, which divides the feature map into several nonoverlapping local windows and only calculates attention within each window

$$\mathbf{A}_{g_s} = \text{Softmax}\left(\frac{\mathbf{Q}_{g_s}\mathbf{K}_{g_s}^T}{\sqrt{d_{g_s}}}\right)\mathbf{V}_{g_s} \tag{17}$$

$$\mathbf{A}_{l_s} = \text{Softmax}\left(\frac{\mathbf{Q}_{l_s}\mathbf{K}_{l_s}^T}{\sqrt{d_{l_s}}}\right)\mathbf{V}_{l_s} \tag{18}$$

where d_{g_s} and d_{l_s} denote the spatial feature dimensions, and the attention matrices $\mathbf{A}_{g_s}$ and $\mathbf{A}_{l_s}$ quantify the association strength between features at different locations within the spatial domain.

After the reshape operation, the feature representations processed by the spatial attention mechanism are obtained. To achieve refined feature selection, this feature is passed through a gating layer to generate the spatial-dimensional gating coefficients $\mathbf{G}_{g_s}$ and $\mathbf{G}_{l_s}$. This gating layer consists of a 1×1 convolution (performing dimension adaptation), a ReLU activation function, a 1×1 convolution (restoring the feature dimension), and a sigmoid function. The value of the gating coefficient is strictly constrained within the interval [0, 1]

$$\langle \mathbf{G}_{g_s}; \mathbf{G}_{l_s} \rangle = \sigma\left(\text{Conv}_{1\times1}\left(\text{Relu}\left(\text{Conv}_{1\times1}\left(\langle \mathbf{A}_{g_s}; \mathbf{A}_{l_s} \rangle\right)\right)\right)\right) \tag{19}$$

where σ denotes the sigmoid function. Then, we perform elementwise multiplication between the two obtained gating coefficients and the features enhanced by CW-SA, respectively. The overall formula can be expressed as follows:

$$\hat{\mathbf{F}}_{g,t_i}^j = \mathbf{G}_{g_s} \odot \mathbf{G}_{g_c} \odot \mathbf{F}_{g,t_i}^j \tag{20}$$

$$\hat{\mathbf{F}}_{l,t_i}^j = \mathbf{G}_{l_s} \odot \mathbf{G}_{l_c} \odot \mathbf{F}_{l,t_i}^j \tag{21}$$

where $\hat{\mathbf{F}}_{g,t_i}^j \in \mathbb{R}^{H/(2^{j+1})\times W/(2^{j+1})\times C''}$ and $\hat{\mathbf{F}}_{l,i}^j \in \mathbb{R}^{H/(2^{j+1})\times W/(2^{j+1})\times C'}$ denote the enhanced global features and enhanced local features, and $\odot$ denotes the Hadamard product. Through this series of operations, the feature responses of key channels can be dynamically enhanced while the interfering information from redundant channels is effectively suppressed and selective optimization in the spatial dimension by dynamically strengthening the features at key spatial locations and suppressing redundant information.

RS images exhibit significant diversity in land cover types. Due to the inherent limitations of local receptive fields in ResNet, the extracted local features may capture detail information irrelevant to the change task, leading to redundancy in the local features. Although the channelwise spatial window attention submodule in the previous stage has performed overall optimization on the features, ResNet, as a local feature extraction network, may still retain some spatial region features with low task relevance in its output. Complementarily, VMamba can generate features focused on global semantic relevance. Directly concatenating the local features extracted by ResNet with the global features extracted by VMamba may cause the discriminability of the fused features to be affected by residual local redundant information. Therefore, we introduce an attention module after the local features processed by the DSJ module, aiming to further suppress redundancy and enhance task-relevant features. Finally, the optimized ResNet features and VMamba features are concatenated along the channel dimension to form a complete feature representation of the two temporal phases. It can be expressed as follows:

$$\mathbf{F}_{t_i}^j = \text{Concat}\left(\text{Att}\left(\hat{\mathbf{F}}_{l,t_i}^j\right), \hat{\mathbf{F}}_{g,t_i}^j\right). \tag{22}$$

C. D^3R Module

The core prerequisite for a model to generate accurate descriptions of image pairs is its ability to effectively distinguish real changes. Previous methods in this field mostly model original features directly via pixel-level differences, which have obvious limitations when analyzing image pairs without real changes. When image pairs exhibit pixel-level differences due to external apparent interference (e.g., temporary occlusion, weather variations), these methods will misidentify such differences as real change signals, prompting the model to classify the pairs as changed ones and trigger the subsequent description generation process. Such misclassification leads the model to acquire incorrect prior knowledge, confuses the feature representations of apparent interference

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and real changes, and ultimately results in a substantial drop in description accuracy.

To address the aforementioned bottleneck, we propose D^{3}R module, as illustrated in Fig. 2. Specifically, we first construct the DIR module by designing a sequential weight generation structure. This module maps the correlation information of bi-temporal features into a dynamic weight tensor with spatial–channel dual relevance. The weight is then applied to the original differences to achieve refined enhancement of pixel-level difference features in both spatial and channel dimensions. Subsequently, we design the change region localization (CRL) module, which adopts a cross-attention mechanism and models the original image features using difference features as guidance signals. While enabling the model to learn difference information, this module establishes an associative relationship between difference features and original image features, avoiding the loss of semantic context of original features caused by learning differences in isolation. Ultimately, it provides high-quality feature support for the subsequent encoder to generate more accurate descriptions. The process can be formulated as follows:

$$\begin{aligned}\mathbf{F}_{\text{fusion}} &= \mathcal{F}_{\text{D}^3\text{R}}\left(\mathbf{F}_{t_0}^j, \mathbf{F}_{t_1}^j\right) \\ &= \mathcal{F}_{\text{CRL}}\left(\mathcal{F}_{\text{DIR}}\left(\mathbf{F}_{t_0}^j, \mathbf{F}_{t_1}^j\right)\right)\end{aligned} \tag{23}$$

where $\mathbf{F}_{\text{fusion}}$ denotes the bi-temporal fused feature highlighting change regions, $\mathbf{F}_{t_0}^j$ and $\mathbf{F}_{t_1}^j$ represent the bi-temporal features extracted through DSJ module, $\mathcal{F}_{\text{DIR}}(\cdot)$ denotes the difference information refinement module, and $\mathcal{F}_{\text{CRL}}(\cdot)$ refers to the change region localization module.

1) Difference Information Refinement: For the DIR module, the input bi-temporal features are denoted as $\mathbf{F}_{t_0}^j \in \mathbb{R}^{L\times B\times D}$ and $\mathbf{F}_{t_1}^j \in \mathbb{R}^{L\times B\times D}$, where L represents the spatial sequence length, B denotes the batch size, and D stands for the number of feature channels. First, $\mathbf{F}_{t_0}^j$ and $\mathbf{F}_{t_1}^j$ are concatenated along the channel dimension to construct a joint feature that contains the correlation information of the two features. Subsequently, a reshape operation is performed on the joint feature to adapt it to the input format $\mathbf{F} \in \mathbb{R}^{LB\times 2D}$ of the weight generation network. To achieve dual-dimensional dynamic modulation, we design a sequential weight generation network $\mathcal{W}_{\text{gen}}$. Through a composite structure consisting of "three linear mappings + two nonlinear ReLU activations + one sigmoid normalization", this network maps $\mathbf{F}$ to a dynamic weight tensor $\mathbf{W} \in \mathbb{R}^{L\times B\times D}$ with dual relevance of the spatial channel. The above operations can be formulated as follows:

$$\mathbf{W} = \text{Reshape}\left(\mathcal{W}_{\text{gen}}\left(\text{Reshape}\left(\text{Concat}\left(\mathbf{F}_{t_0}^j, \mathbf{F}_{t_1}^j\right)\right)\right)\right). \tag{24}$$

This weight tensor $\mathbf{W}$ possesses dual-dimensional modulation capabilities. In the spatial dimension, the values of its elements are dynamically adjusted according to the spatial difference significance between $\mathbf{F}_{t_0}^j$ and $\mathbf{F}_{t_1}^j$. Specifically, it assigns weights approaching $\mathbf{W}_{l,b,d} \to 1$ to target change regions to enhance effective differences and weights approaching $\mathbf{W}_{l,b,d} \to 0$ to background unchanged regions to suppress redundancies. In the channel dimension, it generates independent modulation coefficients for each channel: high weights are assigned to key channels that characterize real changes, while low weights are assigned to redundant channels that reflect apparent interferences.

Subsequently, we apply this dynamic weight tensor to the original difference feature. By performing elementwise multiplication between them, we obtain the enhanced difference feature. This process can be formulated as follows:

$$\hat{\mathbf{F}}_{\text{diff}} = \mathbf{W} \odot \left(\mathbf{F}_{t_1}^j - \mathbf{F}_{t_0}^j\right). \tag{25}$$

Through the above operations, each difference feature value can obtain its exclusive dual-dimensional modulation. This not only effectively enhances the representational capability of difference features but also improves the discriminative accuracy of subsequent tasks.

2) Change Region Localization: For the CRL module, we adopt a cross-attention mechanism. The difference features are used as the keys and values of the cross-attention, and the original image features serve as the queries. This enables the original image features to adaptively focus on the key regions related to differences while preserving the semantic context information of the original features. This can be specifically expressed as follows:

$$\begin{aligned}\hat{\mathbf{F}}_{t_0}^j &= \text{Attn}\left(\mathbf{F}_{t_0}^j, \hat{\mathbf{F}}_{\text{diff}}\right) \\ &= \sigma\left(\frac{\left(\mathbf{F}_{t_0}^j\mathbf{W}^Q\right)\left(\hat{\mathbf{F}}_{\text{diff}}\mathbf{W}^K\right)^T}{\sqrt{d}}\right)\hat{\mathbf{F}}_{\text{diff}}\mathbf{W}^V\end{aligned} \tag{26}$$

$$\begin{aligned}\hat{\mathbf{F}}_{t_1}^j &= \text{Attn}\left(\mathbf{F}_{t_1}^j, \hat{\mathbf{F}}_{\text{diff}}\right) \\ &= \sigma\left(\frac{\left(\mathbf{F}_{t_1}^j\mathbf{W}^Q\right)\left(\hat{\mathbf{F}}_{\text{diff}}\mathbf{W}^K\right)^T}{\sqrt{d}}\right)\hat{\mathbf{F}}_{\text{diff}}\mathbf{W}^V\end{aligned} \tag{27}$$

where $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V$ are the trainable weight matrices.

To avoid the loss of basic semantic information in the original image features caused by relying solely on the attention-updated features, we introduce a residual connection mechanism to fuse the features, thereby generating the final enhanced features $\widehat{\mathbf{F}_{t_0}^j}$ and $\widehat{\mathbf{F}_{t_1}^j}$ for the subsequent subtitle generation task

$$\widehat{\mathbf{F}_{t_0}^j} = \text{LN}\left(\mathbf{F}_{t_0}^j + \text{Dropout}\left(\hat{\mathbf{F}}_{t_0}^j\right)\right) \tag{28}$$

$$\widehat{\mathbf{F}_{t_1}^j} = \text{LN}\left(\mathbf{F}_{t_1}^j + \text{Dropout}\left(\hat{\mathbf{F}}_{t_1}^j\right)\right) \tag{29}$$

where Dropout suppresses model overfitting by randomly masking part of the feature values, while LN performs layer normalization on the fused features to effectively stabilize the feature distribution. This fusion strategy not only preserves the integrity of the semantic context of the original features but also fully integrates the change-region enhancement information guided by the difference features, thereby achieving the dual goal of "semantic preservation-change enhancement." Finally, we perform channel adaptation adjustment on the obtained features $\widehat{\mathbf{F}_{t_0}^j}$ and $\widehat{\mathbf{F}_{t_1}^j}$ through a convolutional layer. Subsequently, we concatenate these features along the channel dimension to construct the final feature $\mathbf{F}_{\text{fusion}}$ that integrates

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the cross-temporal change correlation information. The formula is as follows:

$$\mathbf{F}_{\text{fusion}} = \text{Concat}\left(\text{Conv}\left(\widehat{\mathbf{F}^{\mathbf{j}}_{\mathbf{t_0}}}\right), \text{Conv}\left(\widehat{\mathbf{F}^{\mathbf{j}}_{\mathbf{t_1}}}\right)\right). \tag{30}$$

D. Caption Decoder

The caption decoder is a key component for generating change captions. In this article, we adopt the standard transformer decoder structure, which is composed of multiple stacked decoder layers. Each decoder layer contains three core sublayers: multihead masked self-attention mechanism, encoder–decoder attention mechanism [encoder—decoder attention (EDAtt)], and feed-forward neural network. This structural enables the decoder to fully utilize the input features and contextual information during the generation process while maintaining high computational efficiency.

During the training process, the multihead masked self-attention mechanism ensures that the prediction of each word depends only on the words that have already been generated through masking operations. This masking mechanism ensures the causality of the generation process, enabling the model to generate semantically coherent sentences step by step. The encoder–decoder attention mechanism directs the decoder's attention to the fused features, which allows the decoder to dynamically extract the most relevant information from the fused features for the current generation content, thereby generating more accurate and input-data-fitting captions. Specifically, the query (Q) of the encoder–decoder attention mechanism comes from the output of the previous masked self-attention sublayer, while the key (K) and value (V) are obtained by linearly transforming the fused features. This process can be represented by the following formula:

$$\text{EDAtt}\left(\mathbf{S}^l, \mathbf{F}_{\text{fusion}}\right) = \text{Concat}\left(\text{head}_1, \ldots, \text{head}_h\right)\mathbf{W}_o \tag{31}$$

where

$$\text{head}_m = \text{CrossAtt}\left(\mathbf{S}^l\mathbf{W}^Q_m, \mathbf{F}_{\text{fusion}}\mathbf{W}^K_m, \mathbf{F}_{\text{fusion}}\mathbf{W}^V_m\right) \tag{32}$$

where $\mathbf{W}^Q_m, \mathbf{W}^K_m, \mathbf{W}^V_m \in \mathbb{R}^{d\times d}$ are the trainable matrices of the mth head, $\mathbf{W}_o \in \mathbb{R}^{hd\times d}$ is a trainable matrix to perform linear projection changing the feature channel dimension, $\mathbf{S}^l \in \mathbb{R}^{L\times d}$ is the sentence embedding from the masked self-attention sublayer, L is the length of the sentence, and h is the head number of the MCA.

E. Loss Function

In this article, the cross-entropy loss is adopted as the loss function. During the training phase, the model is optimized by minimizing this function. The loss function can be expressed in the following form:

$$\text{loss} = -\sum_{t=1}^{L}\log\left(\sum_{k=1}^{K}\tilde{\mathbf{y}}_t^{(k)}\mathbf{s}_t^{(k)}\right) \tag{33}$$

where L represents the length of the reference sentence, and K represents the size of the entire vocabulary. $\tilde{\mathbf{y}}_t = [\tilde{y}_t^{(1)}, \tilde{y}_t^{(2)}, \ldots, \tilde{y}_t^{(K)}]$ is the vector representation of the word at the tth position in the reference caption. If the word at the tth position of the reference caption corresponds to the kth word in the vocabulary, then $\tilde{y}_t^{(k)} = 1$, and the rest are 0. $\mathbf{s}_t = [s_t^{(1)}, s_t^{(2)}, \ldots, s_t^{(K)}]$ is the probability vector of each word in the vocabulary predicted by the model when generating the word at the tth position of the caption, and $s_t^{(k)}$ represents the probability that the model predicts the kth word in the vocabulary to appear at the tth position of the generated caption.

IV. Experiments

A. Data Description

LEVIR-CC [32] is a large-scale dataset for RSICC, which captures temporal image changes of 20 regions in TX, USA over a period of 5–15 years. It contains 10 077 image pairs and 50 385 descriptive captions. All bi-temporal images are uniformly resized to 256 × 256 pixels. To comprehensively capture the change information between images, each image pair is annotated by five independent annotators to generate five differentiated descriptions, finally forming a large-scale annotation collection covering multidimensional change features.

B. Experimental Setup

1) Evaluation Metrics: The CC task requires metrics to quantify the degree of matching between the captions generated by the model and human reference sentences. To ensure the fairness and reliability of the experiments, this article uses four widely used image captioning metrics to evaluate the performance of the proposed method, including BLEU-N (N = 1, 2, 3, 4) [38], ROUGE_L [39], METEOR [40], and CIDEr [41]. Higher metric scores indicate better model performance. Additionally, in accordance with [42], we adopt the average metric S^*_m as follows:

$$S^*_m = \frac{1}{4}\left(\text{BLEU-4} + \text{ROUGE_L} + \text{METEOR} + \text{CIDEr}\right). \tag{34}$$

2) Implementation Details: RMNet is implemented with PyTorch, and all experiments are conducted on an NVIDIA GTX 3090 GPU. Considering the need to balance information integrity and computational efficiency in feature extraction, we select the first three network stages of ResNet and VMamba for feature extraction. During the training phase, we use the AdamW optimizer with the initial learning rates for both the encoder and the decoder set to $2e^{-4}$. We set the maximum number of epochs to 40 and the batch size to 64. After each epoch, we evaluate the model's performance on the validation set. If the BLEU-4 score decreases for three consecutive epochs, the learning rate is decayed by a factor of 0.7. The best model on the validation set is selected to be evaluated on the test set, and the average of five results is taken as the final result.

3) Comparison With the State-of-the-Art Methods: In this section, to systematically evaluate the performance of the proposed model, we conduct a comparative analysis with several state-of-the-arts methods in the current field. Capt-Rep-Diff [16], Capt-Att [16], Capt-Dual-Att [16], and DUDA [16]

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TABLE I

QUANTITATIVE ASSESSMENT OF DIFFERENT CC METHODS ON LEVIR-CC DATASET. CNN‡ INDICATES THAT ONLY THE RESNET101 IS USED IN THE FEATURE EXTRACTION STAGE, WHILE TRANSFORMER‡ MEANS ONLY THE VIT_B_16 IS ADOPTED IN THIS STAGE; THE REMAINING STAGES ARE CONSISTENT WITH THE PROPOSED METHOD RMNET (COLOR CONVENTION: BEST AND 2$_{ND\text{-}BEST}$)

Type	Method	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE$_L$	CIDEr	$\mathbf{S}_m^*$	Param.
CNN based	Capt-Rep-Diff	0.7290	0.6198	0.5362	0.4741	0.3447	0.6564	1.1057	0.6452	73.21M
	Capt-Att	0.7764	0.6740	0.5924	0.5315	0.3658	0.6973	1.2122	0.7017	73.60M
	Capt-Dual-Att	0.7951	0.7057	0.6323	0.5746	0.3656	0.7069	1.2442	0.7228	75.58M
	DUDA	0.8144	0.7222	0.6424	0.5779	0.3715	0.7104	1.2432	0.7258	80.31M
Transformer based	MCCFormers-D	0.7860	0.6934	0.6171	0.5553	0.3598	0.6871	1.2264	0.7071	68.98M
	MCCFormers-S	0.7945	0.7164	0.6454	0.5907	0.3653	0.6984	1.2206	0.7187	69.86M
	RSICCformer	0.8255	0.7437	0.6751	0.6196	0.3829	0.7178	1.2949	0.7538	56.22M
	PSNet	0.8386	0.7513	0.6789	0.6211	0.3880	0.7360	1.3262	0.7678	424.39M
	PromptCC	0.8277	0.7529	0.6886	0.6323	0.3846	0.7409	1.3526	0.7776	872.70M
	SEN	0.8381	0.7660	0.7039	0.6516	0.3931	0.7424	1.3517	0.7847	39.86M
Mamba based	RSCaMa	0.8579	0.7799	0.7104	0.6524	0.3991	0.7524	1.3656	0.7924	790.18M
-	CNN‡	0.8257	0.7414	0.6703	0.6109	0.3926	0.7316	1.3233	0.7646	97.46M
	Transformer‡	0.8492	0.7648	0.6914	0.6294	0.3941	0.7467	1.3741	0.7860	151.94M
-	**RMNet**	**0.8609**	**0.7843**	**0.7168**	**0.6611**	**0.4094**	**0.7557**	**1.3943**	**0.8051**	148.18M

employ a hybrid framework of CNNs and attention mechanisms for bi-temporal feature processing, with LSTM utilized for language decoding. MCCFormer-S [43], MCCFormer-D [43], RSICCFormer [32], PSNet [28], PromptCC [33], and SEN [44] all take transformer as the core component for constructing the neck structure of the caption generation network and adopt the transformer decoder to complete the generation process of variable captions. RSCaMa [20] method performs feature extraction operations based on the mamba model.

C. Results and Analysis

1) Quantitative Results and Analysis: As shown in Table I, it simultaneously presents the performance metrics and parameter overhead of various methods on the LEVIR-CC dataset. It can be seen that compared with the state-of-the-art RSCaMa, RMNet improves by 0.87%, 1.03%, 0.33%, and 2.87% points in BLEU-4, METEOR, ROUGE_L, and CIDEr, respectively. The comprehensive index S_m^* increases by 1.27% points. In terms of parameter count, RSCaMa has a parameter count as high as 790.18 M while RMNet's is only 148.18 M, representing a reduction of approximately 81.2% in parameter count. This significantly reduces computational overhead while achieving substantial performance improvements. For the variant CNN‡ and transformer‡, there is also a noticeable performance gap compared to RMNet. The transformer‡ has a parameter count of 151.94 M, which is close to that of RMNet, yet its BLEU-4 score is only 0.6294 whereas RMNet boosts this metric to 0.6611. While the CNN‡ has a relatively low parameter count of 97.46 M, its comprehensive metric $\mathbf{S}_m^*$ is only 0.7646, which is significantly lower than RMNet's 0.8051. This fully demonstrates that the CNN-Mamba hybrid architecture adopted in RMNet can significantly improve the model's comprehensive performance on the premise of effectively controlling the parameter count.

The performance breakthrough of RMNet is attributed to its dual-branch structure fusing VMamba and ResNet, which enables the model to accurately capture spatial local details while efficiently modeling global feature correlations, thereby realizing the synergistic utilization of local–global information. Meanwhile, the difference weight matrix mechanism suppresses pseudo-change information caused by factors, such as illumination and noise, strengthens the feature response to real change regions, and thus improves the accuracy of change characterization. Compared with other methods, RMNet achieves the highest scores in all metrics—reaching 66.11% BLEU-4, 40.94% METEOR, 75.57% ROUGE_L, and 139.43% CIDEr, which further verifies its significant advantages in the task of RS image change caption generation.

2) Qualitative Results and Analysis: To more intuitively demonstrate the effectiveness of the proposed method, we compare the caption generation results between RMNet and the previous state-of-the-art methods, including MCCFormer-S, MCCFormer-D, RSICCFormer, and SEN, and conduct a qualitative analysis. A total of six representative image pairs are selected, consisting of four pairs of changed images and two pairs of unchanged images. As illustrated in Fig. 4, MCCFormer-S performs poorly in the CC task, exhibiting pronounced lexical redundancy that leads to insufficient descriptive accuracy. For image pairs containing real changes, MCCFormer-D demonstrates relatively strong capabilities in change region detection and description generation, achieving a high degree of visual–textual alignment. However, its robustness is limited when handling image pairs without actual changes, resulting in a high false-positive rate. For example, in the fifth and sixth groups of unchanged image pairs, the generated descriptions incorrectly indicate scene changes. The primary cause of such misjudgments lies in pseudo-change information introduced by illumination variations, which interferes with the model's feature extraction

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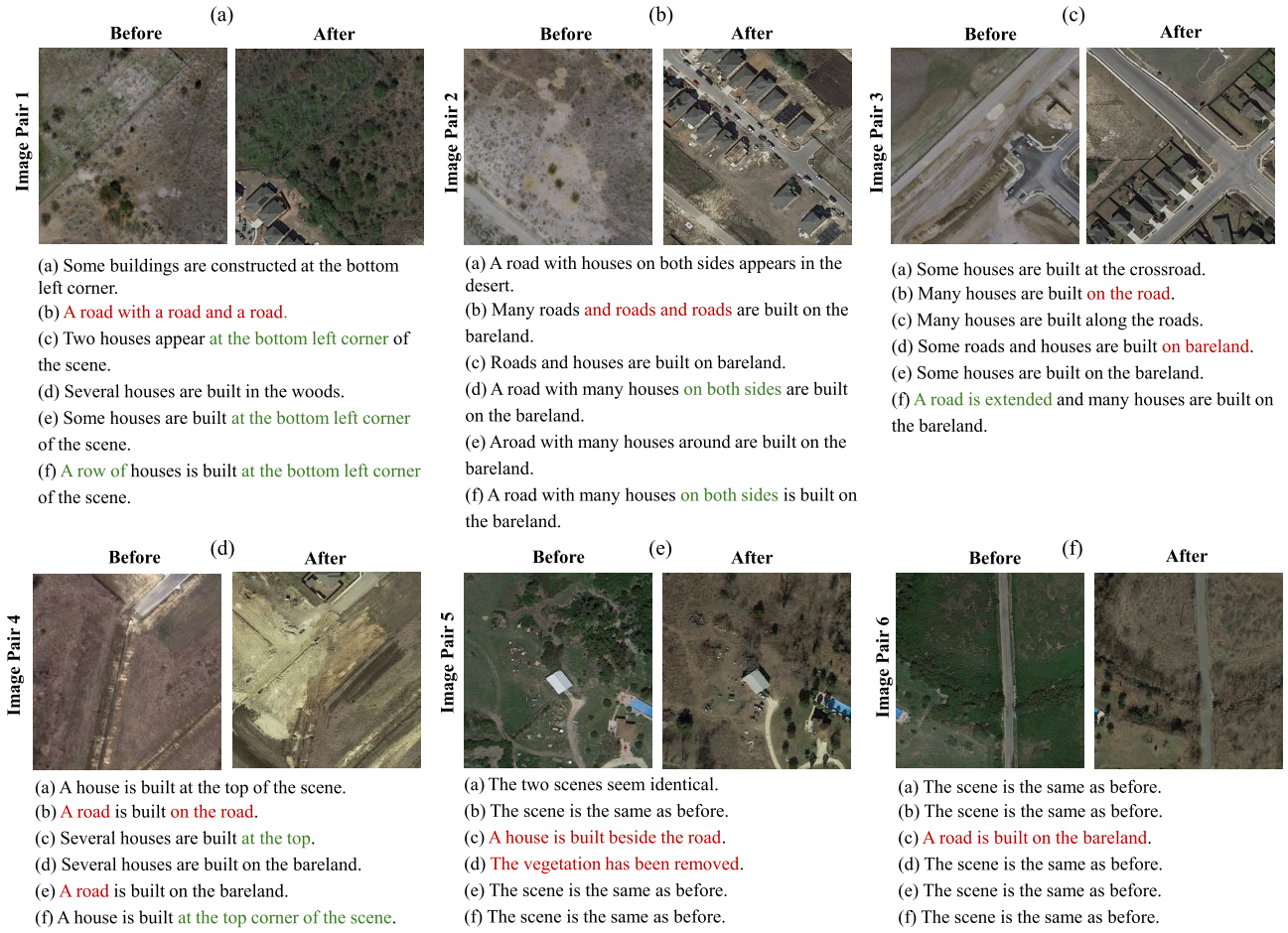


Fig. 4. Qualitative comparison of captioning results of different methods on the LEVIR-CC dataset. (a) One of the five ground-truth captions. (b) MCCFormer-S. (c) MCCFormer-D. (d) RSICCFormer. (e) SEN. (f) RMNet. More accurate and detailed words are marked in green. Red words are not accurate.

and change discrimination modules, causing pixel-level intensity differences to be mistakenly interpreted as real changes. RSICCFormer suffers from a similar issue.

Compared with the aforementioned models, SEN achieves a notable improvement in description accuracy; however, its generated captions remain overly coarse. While SEN can summarize the existence of changes, it fails to precisely localize the changed objects in space, resulting in insufficient spatial specificity in the descriptions. In contrast, RMNet achieves the best overall performance among all compared models. It can accurately distinguish real scene changes while effectively suppressing pseudo-change interference, and it clearly identifies the spatial locations of changed objects in the generated descriptions, thereby realizing a complete closed loop of "change detection–region localization–precise description."

Fig. 5 shows the attention weight visualizations of DA and EDA for image pairs with real changes. When DA applies difference information to original image features only via simple pixel difference, its attention weight allocation is flawed: it often guides the model to focus on false change features from nontarget factors (e.g., illumination fluctuations, random noise) instead of real change regions. Take the third image pair in the figure as an example. The core real change in this image pair is a newly added building in the lower-left area, but DA not only assigns weights to this core region but also gives high weights to surrounding irrelevant regions. This unreasonable allocation causes information bias in the model's subsequent change descriptions, which prevents accurate targeting of the core change and reduces description accuracy.

In contrast, the EDA optimized by the difference enhancement strategy exhibits better performance in attention weight allocation. Through the effects of enhancing real change features and suppressing false change features, EDA can accurately assign high attention weights to the core real change regions that need to be emphasized in descriptions, while effectively avoiding the interference of irrelevant regions on description generation. This enables the model to focus on the attribute and spatial information of the core change target when generating change captions, and the output descriptions have stronger directivity, significantly improving the accuracy of change captions.

D. Ablation Study and Analysis

1) Effectiveness of DSJ Feature Extraction: To fully leverage the complementary advantages of ResNet and VMamba, we propose a joint feature extraction module. This module integrates local feature extraction capability of ResNet with

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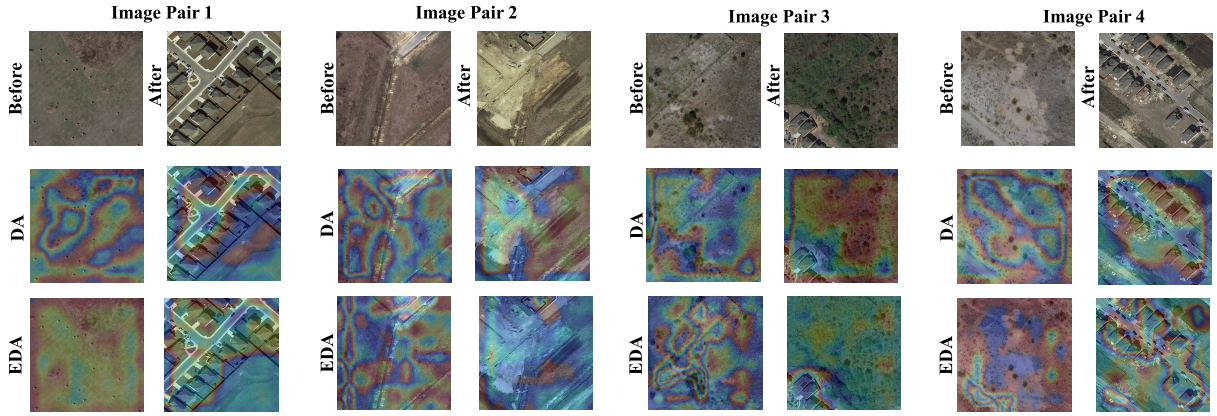


Fig. 5. Attention weight visualization of DA and EDA for image pairs with real changes. DA applies difference information to the original image features only through simple pixel differences; EDA acts on the original image features using the dual-dimension enhanced differences derived from the proposed method.

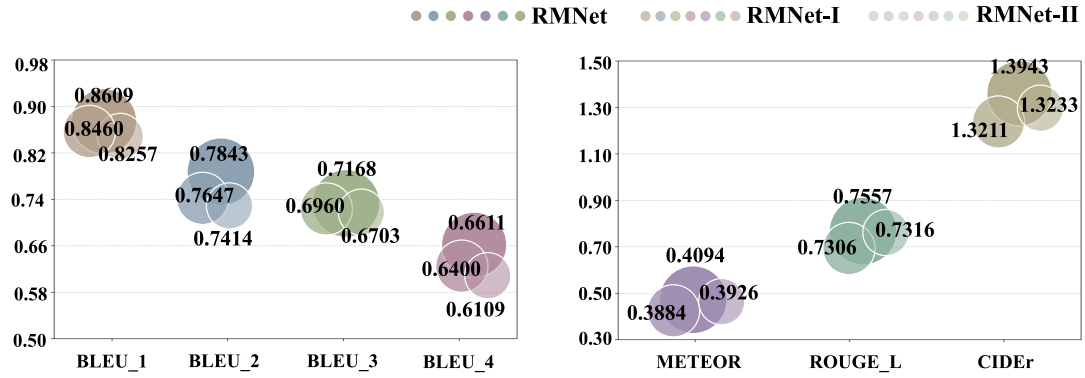


Fig. 6. Performance comparison of RMNet, RMNet-I (only VMamba), and RMNet-II (only ResNet) on the LEVIR-CC dataset.

global context modeling ability of VMamba, aiming to comprehensively extract the local fine-grained features and global semantic context features of RS images. These fused features can provide more comprehensive and accurate feature support for downstream change caption generation tasks. To verify the effectiveness of this module, two variants are designed: RMNet-I (only VMamba) and RMNet-II (only ResNet). Fig. 6 presents the performance comparison of RMNet, RMNet-I, and RMNet-II in the RS C task. It is evident that the proposed RMNet achieves the optimal performance across all evaluation metrics. This validates the technical advantage of fusing ResNet and VMamba for joint feature extraction and simultaneously confirms that the complementary integration of local and global features plays a crucial role in improving the performance of the RS CC task.

2) Impact of Channelwise Spatial Window Attention: After completing the basic feature extraction, we design a channelwise spatial window attention submodule to achieve accurate screening and enhancement of key features. This module not only avoids the inefficiency of global computation in traditional spatial attention but also makes up for the deficiency of traditional channel attention in capturing global correlations. To verify the effectiveness of the submodule from objective results, we design a variant model RMNet-III,

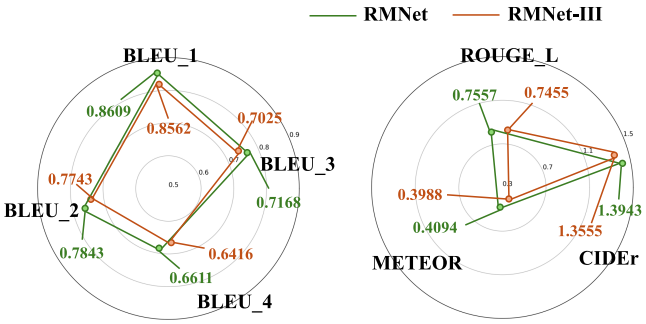


Fig. 7. Performance comparison under different spatial–channel attention mechanisms.

which adopts the channel attention mechanism and spatial attention mechanism proposed by the CBAM [45] module for feature selection. Fig. 7 presents the performance comparison results between RMNet and RMNet-III on the LEVIR-CC dataset. Experimental results show that RMNet outperforms RMNet-III across all evaluation metrics: BLEU-4 improves by nearly 2% and CIDEr by nearly 4%. Moreover, when the channel dimension is 1024, the independent parameter count of the channelwise attention submodule is 3.84 M, accounting for only 2.59% of the model's total parameters. These data

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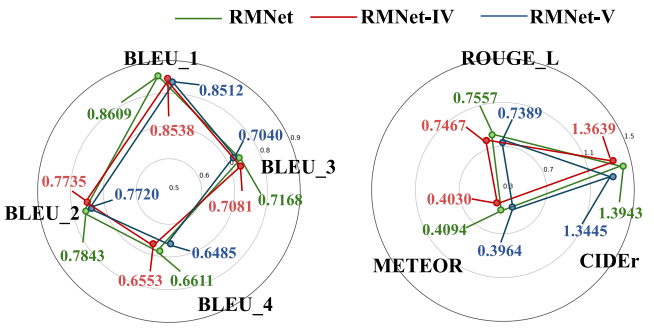


Fig. 8. Performance comparison under different difference enhancement strategies.

fully demonstrate that this submodule effectively enhances the ability to select and represent key features without significantly increasing computational cost, thereby achieving efficient feature optimization.

3) Efficacy of DIR: In the process of differential feature processing, directly applying pixel-level difference information to original image features tends to cause the model to assign excessively high attention weights to pseudo-change regions induced by nontarget factors. This not only interferes with the effective extraction of real change features but also degrades the performance of downstream tasks. To address this issue, we design a D^{3}R module. This module implements differential information enhancement strategies, respectively, in the spatial dimension and channel dimension, thereby achieving the feature optimization goal of highlighting core change regions and suppressing irrelevant pseudo-changes.

The previous attention weight visualization results in Fig. 5 have verified the effectiveness of the D^{3}R module from a qualitative perspective. To further reveal the module's mechanism of action intuitively from a quantitative perspective, we design two comparative variants: RMNet-IV (without the channel-dimensional enhancement mechanism) and RMNet-V (without any enhancement processing). Fig. 8 presents the performance comparison results of RMNet, RMNet-IV, and RMNet-V in relevant tasks. Experimental data show that RMNet-V, which does not perform any enhancement on differential features, achieves the worst performance, while the RMNet method integrating spatial–channel dual-dimensional enhancement achieves the optimal performance across all evaluation metrics. This empirical result fully confirms that the strategy of synchronously implementing spatial-dimensional and channel-dimensional enhancement on differential features can effectively improve the representation ability of real change features, significantly suppress pseudo-change interference, and provide key support for model performance optimization.

V. Conclusion

In this article, we propose RMNet for RSICC. It innovatively adopts a DSJ feature extraction and fusion architecture and a D^{3}R mechanism, which specifically addresses current task limitations and enhances generalization in complex scenarios. To solve insufficient feature extraction of single-stream architectures, DSJ module is designed. Via cross-network feature interaction between VMamba and ResNet, it fully extracts global semantic and local detail information from bi-temporal RS images. To resolve false change detection errors caused by pixel-level differences, the D^{3}R module is constructed, which refines difference features in spatial and channel dimensions, suppressing redundant interference and building an accurate change pattern spectrum. Finally, spatially interpretable change descriptions are generated in the transformer decoder. Experiments on the LEVIR-CC dataset show that RMNet outperforms mainstream methods in all metrics, verifying its effectiveness in overcoming RSICC limitations.

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